TECHNICKÁ UNIVERZITA V KOŠICIACH

FAKULTA ELEKTROTECHNIKY A INFORMATIKY

Problem Set 6
NEON VENDETTA

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Assignment

Using the ncurses library, a 2D action game Neon Vendetta 2025 was created. The project fulfills all required criteria:

- 2D world represented as a character grid
- Work with terminal colors
- Real-time keyboard control (no Enter needed)
- Multiple handcrafted levels
- Time-based game mechanics
- Project complexity exceeding basic examples

This game includes enemy AI, combo mechanics, particle effects and a full UI panel.

Game Logic

The code consists of multiple functional blocks:

- init_curses() – initializes ncurses, colors, input
- init_level() – loads one of two level maps
- is_solid() – collision detection
- attack_enemy() – kills enemies and spawns blood particles
- player_attack() – melee & pistol logic + LOS detection
- update_ai() – enemy movement, detection & game over
- update_game() – timers, combos, particles, AI update
- draw_world() – renders environment, player, enemies, blood
- draw_menu(), draw_help(), draw_gameover(), draw_victory() – UI screens

World Description

The world is constructed using a 2D grid. Elements include:

Dynamic elements:
• '@' – player
• 'E' – enemy
• 'A' – ammo pickup
• blood particles and corpse markers

Static elements:
• '#' – wall (blocking)
• '.' – floor
• '>' – exit



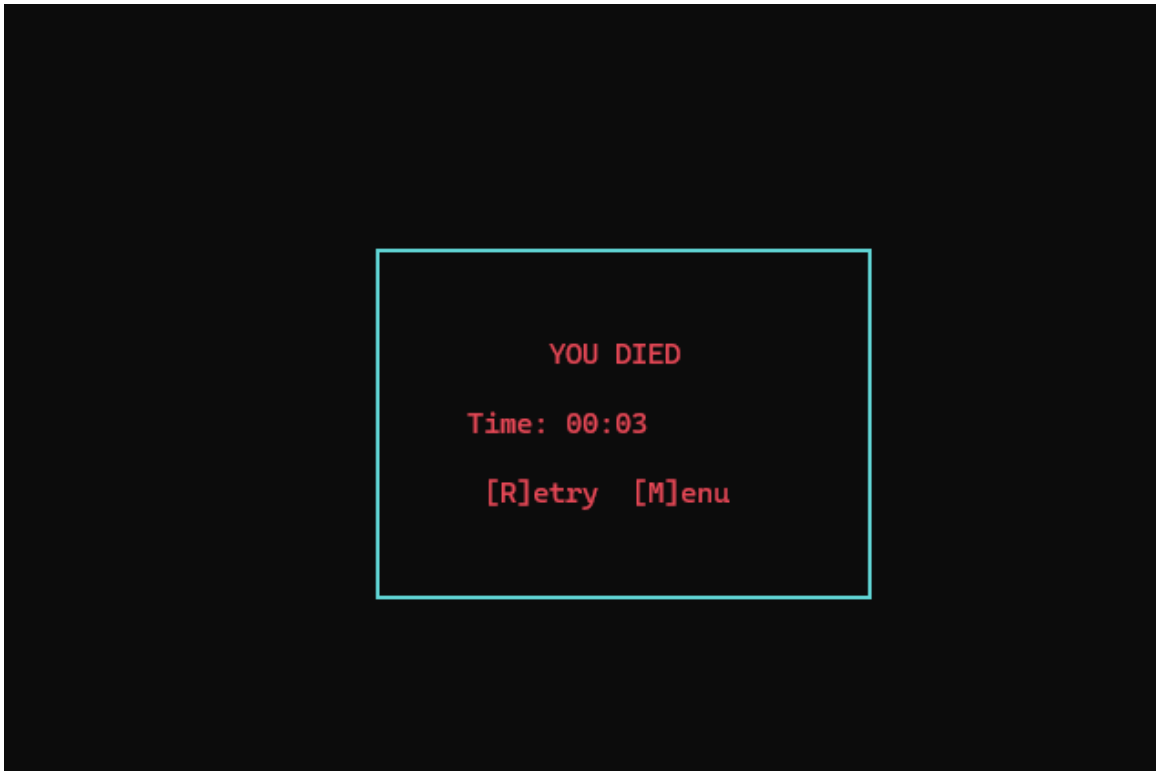
Game Play

This game is not a remake of any known title. The player moves inside a neon-styled terminal level, eliminates enemies, and reaches the exit.

Victory conditions:
1. Eliminate all enemies
2. Reach the exit '>'

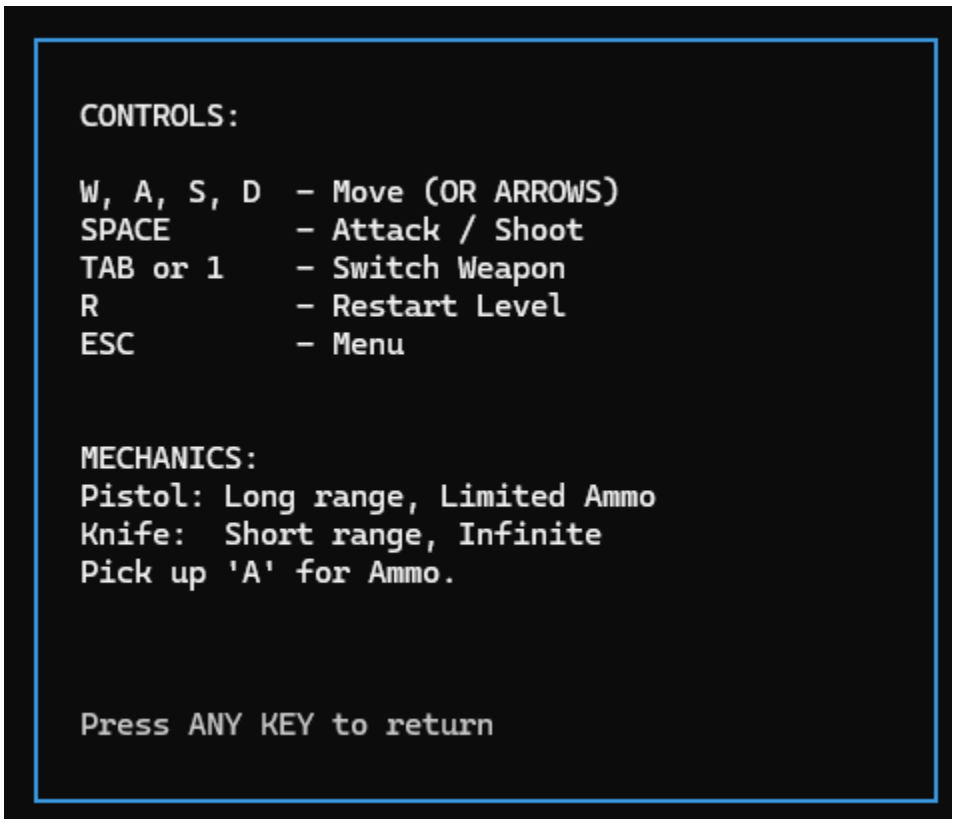
Weapons:
• Knife – infinite melee
• Pistol – ranged, uses ammo

If the player is touched by an enemy, the game ends. After clearing a level, the player transitions to the next.



Controls

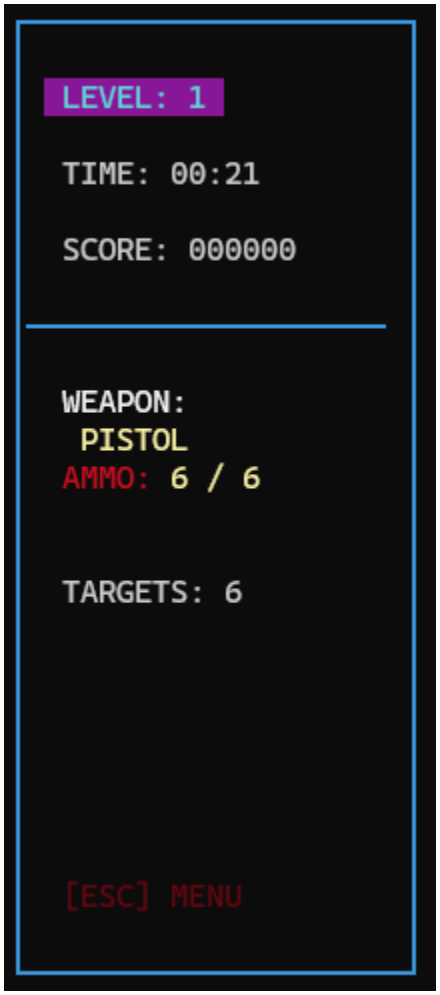
• Movement: W A S D / Arrow keys
• Attack: SPACE
• Switch weapon: TAB / 1
• Restart level: R
• Menu: ESC



User Interface

The right panel shows:

- Level number
- Timer (MM:SS)
- Score
- Weapon + ammo
- Remaining enemies
- Combo multiplier
- EXIT OPEN indicator



Conclusion

Neon Vendetta demonstrates a fully functional ncurses-based action game. It features AI, scoring, particles, a clean UI, and multiple levels. The project may be further expanded with more levels, refined AI or graphical versions.

